

Python: warm-up exercises on iteration

This notebook was generated in student mode.

Exercise: Coin Flipping Simulation

Exercise Description

Write a piece of code that simulates coin tosses and keeps track of heads and tails.

Requirements:

- Use a variable `n` to specify the number of coin tosses
- Keep track of the number of heads and tails
- Use `random.randint(1, 100)` to simulate coin tosses (>50 = heads, ≤ 50 = tails)
- Store the final counts in variables `heads` and `tails`
- Return the variables `heads` and `tails`

Your solution

```
import random

def simulate_coin_flips(n):
    """Simulate n coin tosses and return (heads, tails).
    Each toss uses random.randint(1, 100) (>50=heads).
    """
    # TODO: implement loop and counts
    pass
```

Test your solution

```
# Test that heads + tails equals total number of tosses
n = 10
heads, tails = simulate_coin_flips(n)
total_tosses = heads + tails
assert total_tosses == n, f"Expected {n} total tosses, got {total_tosses}"
assert heads >= 0 and tails >= 0, 'Counts should be non-negative'
print('Test passed: coin flipping simulation works correctly!')
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pythontutor.'
```

Exercise: Sum of Number Sequence

Exercise Description

Write a piece of code that calculates the sum of a sequence of numbers.

Requirements:

- Keep track of the sum of numbers processed so far
- Stop when you encounter 0 (don't include 0 in the sum)
- Store the final sum in a variable called `total_sum`
- Return `total_sum`

Your solution

```
def sum_until_zero(numbers):  
    """Return the sum of numbers until the first 0 (excluded).  
    The input is a list of integers.  
    """  
    # TODO: iterate and stop at 0, accumulate sum  
    pass
```

Test your solution

```
numbers = [5, 3, 8, 2, 0]  
total_sum = sum_until_zero(numbers)  
expected_sum = 18  
assert total_sum == expected_sum, f"Expected sum {expected_sum}, got {total_sum}"  
print('Test passed: sum calculation works correctly!')
```

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```

Exercise: Count Odd and Even Numbers

Exercise Description

Write a piece of code that counts the number of odd and even numbers in a given range.

Requirements:

- Count numbers in the range from 1 to n (inclusive)
- Count how many numbers are odd and how many are even
- Store the counts in variables `odd_count` and `even_count`
- Return the variables `odd_count` and `even_count`

Your solution

```
def count_odd_even(n):  
    """Return a tuple (odd_count, even_count) for numbers 1..n.  
    """  
    # TODO: loop and count by parity  
    pass
```

Test your solution

```
n = 10  
odd_count, even_count = count_odd_even(n)  
assert odd_count == 5, f"Expected 5 odd numbers, got {odd_count}"  
assert even_count == 5, f"Expected 5 even numbers, got {even_count}"
```

```
assert odd_count + even_count == n, f"Total count should equal {n}"  
print('Test passed: odd/even counting works correctly!')
```

Debug your solution

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```

Exercise: Generate Fibonacci Sequence

Exercise Description

Write a piece of code that generates the first n numbers of the Fibonacci sequence.

Requirements:

- Generate the first 7 Fibonacci numbers
- The Fibonacci sequence starts with 0 and 1, then each subsequent number is the sum of the two preceding ones
- Store all the generated numbers in a list called `fibonacci_numbers`
- For `n=7`, the sequence should be: [0, 1, 1, 2, 3, 5, 8]
- Return the variable `fibonacci_numbers`

Definition: Each number in the Fibonacci sequence is given by the sum of the two preceding ones. \The Fibonacci sequence starts with two 'seed' numbers 0 and 1) and then the proper sequence starts: 0, 1, 1, 2, 3, 5, 8, 13, 21, ...

Your solution

```
def fibonacci(n):  
    """Return list with first n Fibonacci numbers starting from 0, 1.  
    """  
    # TODO: build the sequence of length n  
    pass
```

Test your solution

```
n = 7  
fibonacci_numbers = fibonacci(n)  
expected_sequence = [0, 1, 1, 2, 3, 5, 8]  
assert fibonacci_numbers == expected_sequence, f"Expected {expected_sequence}, got {fibonacci
```

```
assert len(fibonacci_numbers) == n, f"Expected {n} numbers, got {len(fibonacci_numbers)}"  
print('Test passed: Fibonacci sequence generation works correctly!')
```

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```

Exercise: Multiples of 3 and 5

Exercise Description

Write a piece of code that processes numbers from 1 to n and creates a result list based on multiples.

Requirements:

- Process integers from 1 to n
- For multiples of both 3 and 5, add 'Mult35' to the result
- For multiples of 3 only, add 'Mult3' to the result
- For multiples of 5 only, add 'Mult5' to the result
- For other numbers, add the number itself to the result
- Store all results in a list called `result` and return it

Your solution

```
def classify_multiples(n):  
    """Return a list for numbers 1..n with 'Mult35'/'Mult3'/'Mult5' or the number.  
    """  
    # TODO: iterate and append appropriate value  
    pass
```

Test your solution

```
n = 15  
result = classify_multiples(n)  
assert result[14] == 'Mult35', f"Expected 'Mult35' at position 14, got {result[14]}"  
assert result[2] == 'Mult3', f"Expected 'Mult3' at position 2, got {result[2]}"  
assert result[4] == 'Mult5', f"Expected 'Mult5' at position 4, got {result[4]}"  
assert len(result) == n, f"Expected {n} elements, got {len(result)}"  
print('Test passed: multiples classification works correctly!')
```

Debug your solution

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```

Exercise: Print Number Subset

Exercise Description

Write a piece of code that scans through a range of numbers but only collects a subset.

Requirements:

- Scan all integers from 1 to n
- Only collect numbers from 1 to m (where $m \leq n$)
- Store the collected numbers in a list called `subset_numbers` and return it

Your solution

```
def collect_subset(n, m):  
    """Return list of integers from 1..m while scanning 1..n.  
    Precondition: m <= n.  
    """
```

```
# TODO: build and return the subset list
pass
```

Test your solution

```
n = 10
m = 6
subset_numbers = collect_subset(n, m)
expected_subset = [1, 2, 3, 4, 5, 6]
assert subset_numbers == expected_subset, f"Expected {expected_subset}, got {subset_numbers}"
assert len(subset_numbers) == m, f"Expected {m} numbers, got {len(subset_numbers)}"
assert max(subset_numbers) == m, f"Expected max value {m}, got {max(subset_numbers)}"
print('Test passed: subset collection works correctly!')
```

Debug your solution

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```

Exercise: Find Prime Numbers

Exercise Description

Write a piece of code that finds all prime numbers smaller than a given number n .

Requirements:

- Find all prime numbers smaller than n
- A prime number is greater than 1 and has no divisors other than 1 and itself
- Store all found prime numbers in a list called `prime_numbers` and return it
- Use a simple primality test: check if a number is divisible by any number from 2 to $n/2$

Your solution

```
def primes_less_than(n):  
    """Return a list of all prime numbers strictly less than n.  
    """  
    # TODO: nested loop primality test and collect primes  
    pass
```

Test your solution

```
n = 15  
prime_numbers = primes_less_than(n)  
expected_primes = [2, 3, 5, 7, 11, 13]  
assert prime_numbers == expected_primes, f"Expected {expected_primes}, got {prime_numbers}"  
assert all(p < n for p in prime_numbers), 'All primes should be less than n'  
print('Test passed: prime number finding works correctly!')
```

Debug your solution

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```

Exercise: Find Pythagorean Triples

Exercise Description

Write a piece of code that finds all Pythagorean triples for a given number n .

Requirements:

- Find all pairs (x, y) such that $x^2 + y^2 = n^2$
- Both x and y should be positive integers less than n
- Store all valid pairs as tuples in a list called `pythagorean_pairs` and return it
- A Pythagorean triple consists of three positive integers x , y , and n such that $x^2 + y^2 = n^2$

Your solution

